

Day: _____

HW 4th & 5th Date: _____

Assignment

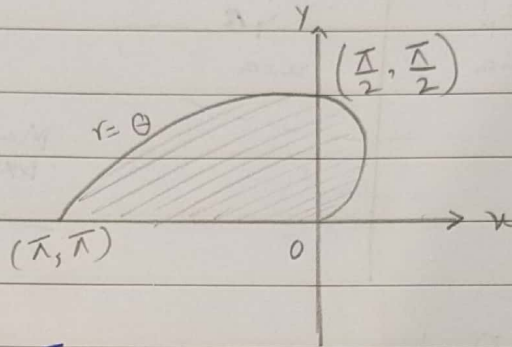
Exercise 11.5

$$\cos 2\theta = 1 - 2\sin^2\theta$$

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

1. Bounded by the spiral
- $r = \theta$
- for
- $0 \leq \theta \leq \pi$

$$\text{Area} = \int \frac{1}{2} r^2 d\theta$$



$$= \int_0^{\pi} \frac{1}{2} \theta^2 d\theta = \frac{1}{2} \int_0^{\pi} \theta^2 d\theta = \frac{1}{2} \left| \frac{\theta^3}{3} \right|_0^{\pi}$$

$$= \frac{1}{2} \left[\frac{\pi^3}{3} - \frac{0^3}{3} \right] = \frac{1}{2} \left| \frac{\pi^3}{3} \right| = \boxed{\frac{\pi^3}{6}}$$

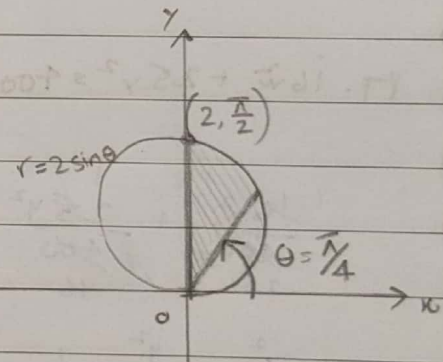
2. Bounded by the circle
- $r = 2\sin\theta$
- for
- $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$

$$A = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{2} [2\sin\theta]^2 d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{2} [4\sin^2\theta] d\theta$$

$$= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2\theta d\theta = 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{2} - \frac{\cos 2\theta}{2} d\theta = 2 \left| \frac{\theta}{2} - \frac{\sin 2\theta}{4} \right|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 2 \left[\left(\frac{\frac{\pi}{2}}{2} - \frac{\sin 2(\frac{\pi}{2})}{4} \right) - \left(\frac{\frac{\pi}{4}}{2} - \frac{\sin 2(\frac{\pi}{4})}{4} \right) \right]$$

$$= 2 \left| \frac{\pi}{4} - 0.142699 \right| = \boxed{1.28}$$



Day: _____

Date: _____

3. Inside the oval Limaçon $r = 4 + 2 \cos \theta$

$$A = \frac{1}{2} \int_0^{2\pi} [4 + 2 \cos \theta]^2 d\theta = \frac{1}{2} \int_0^{2\pi} 16 + 16 \cos \theta + 4 \cos^2 \theta d\theta$$

$$= \int_0^{2\pi} 8 + 8 \cos \theta + 2 \cos^2 \theta d\theta = \left[8\theta + 8 \sin \theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\frac{\cos 2\theta + 1}{2} = \cos^2 \theta$$

$$= \int_0^{2\pi} 8 + 8 \cos \theta + 2 \left[\frac{\cos 2\theta + 1}{2} \right] d\theta$$

$$= \left[8\theta + 8 \sin \theta + \frac{\sin 2\theta}{2} + \theta \right]_0^{2\pi} = \left[8(2\pi) + 8 \sin(2\pi) + \frac{\sin 2(2\pi)}{2} + 2\pi \right]$$

$$= 16\pi + 0 + 0 + 2\pi = \boxed{18\pi}$$

4. Inside the cardioid $r = a(1 + \cos \theta)$, $a > 0$

$$A = \int_0^{2\pi} \frac{1}{2} [a(1 + \cos \theta)]^2 d\theta = \frac{1}{2} a^2 \int_0^{2\pi} 1 + 2 \cos \theta + \cos^2 \theta d\theta$$

$$= \frac{1}{2} a^2 \int_0^{2\pi} 1 + 2 \cos \theta + \frac{\cos 2\theta + 1}{2} d\theta = \frac{1}{2} a^2 \left[\theta + 2 \sin \theta + \frac{\sin 2\theta}{2} + \frac{\theta}{2} \right]_0^{2\pi}$$

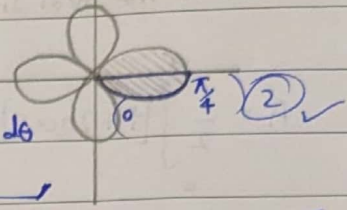
$$= \frac{1}{2} a^2 \left[\left(2\pi + 2 \sin(2\pi) + \frac{\sin 2(2\pi)}{2} + \frac{2\pi}{2} \right) - \left(0 + 2 \sin 0 + \frac{\sin 2(0)}{2} + \frac{0}{2} \right) \right]$$

$$= \boxed{\frac{3}{2} \pi a^2}$$

Day: _____

Date: _____

5. Inside one leaf of the four-leaved rose $r = \cos 2\theta$



$$A = 2 \int_0^{\pi/4} \frac{1}{2} \cos^2 2\theta \, d\theta = \int_0^{\pi/4} \cos 2\theta \cdot \cos 2\theta \, d\theta \quad (\text{or}) \quad \int_0^{\pi/4} \cos^2 2\theta \, d\theta$$

$$= \int_0^{\pi/4} \frac{\cos 2u}{2} \, du = \frac{1}{2} \int_0^{\pi/4} \frac{\cos 2u}{2} \, du$$

$$= \frac{1}{2} \left[\frac{\sin 2u}{4} + \frac{u}{2} \right]_0^{\pi/4}$$

$$\begin{aligned} \cos 2\theta &= u \\ du &= -2 \sin 2\theta \, d\theta \\ d\theta &= -\frac{1}{2 \sin 2\theta} du \end{aligned}$$

$$\begin{aligned} du &= -2 \cos \theta \\ du &= 2 \sin \theta \cos \theta (-2) \, d\theta \end{aligned}$$

$$\frac{du}{-4 \sin \theta \cos \theta} = d\theta$$

Using substitution method

take $2\theta = u$ new limits

$$du = 2 \, d\theta \quad \hookrightarrow \quad 2\left(\frac{\pi}{4}\right) = u \Rightarrow \frac{\pi}{2} \checkmark$$

$$\frac{1}{2} \cdot du = d\theta$$

$$2(0) = u \Rightarrow 0 \checkmark$$

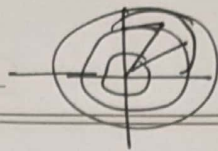
$$\begin{aligned} 2\theta &= u \\ du &= 2 \, d\theta \\ du \cdot \frac{1}{2} &= d\theta \end{aligned}$$

$$= \int_0^{\pi/2} \frac{1}{2} \cos^2 u \, du = \frac{1}{2} \int_0^{\pi/2} \frac{\cos 2u + 1}{2} \, du = \frac{1}{2} \left[\frac{\sin 2u}{4} + \frac{u}{2} \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\frac{\sin 2\left(\frac{\pi}{2}\right) + \frac{\pi}{2}}{4} \right] \Rightarrow \frac{1}{2} \left[0 + \frac{\pi}{4} \right] = \boxed{\frac{\pi}{8}}$$

0 →

Day:

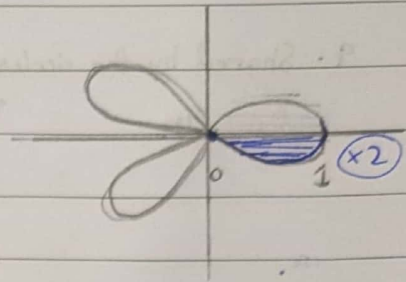


Date:

6. Inside one leaf of the three-leaved rose $r = \cos 3\theta$

$$A = 2 \int_0^{\frac{\pi}{6}} \frac{1}{2} \cos^2 3\theta \, d\theta$$

$$\begin{aligned} 3\theta &= \frac{\pi}{2} \\ \theta &= \frac{\pi}{6} \end{aligned} \quad \boxed{\theta = 0}$$



using substitution method

$$3\theta = u$$

$$du = 3 \, d\theta$$

$$\boxed{d\theta = \frac{1}{3} \cdot du}$$

changing the limits

$$u = 3(0)$$

$$\boxed{u = 0}$$

$$u = 3\left(\frac{\pi}{6}\right)$$

$$\boxed{u = \frac{\pi}{2}}$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{3} \cos^2 u \, du = \frac{1}{3} \int_0^{\frac{\pi}{2}} \frac{\cos 2u + 1}{2} \, du = \frac{1}{3} \left[\frac{\sin 2u}{4} + \frac{u}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{3} \left[\frac{\sin 2\left(\frac{\pi}{2}\right)}{4} + \frac{\frac{\pi}{2}}{2} \right] =$$

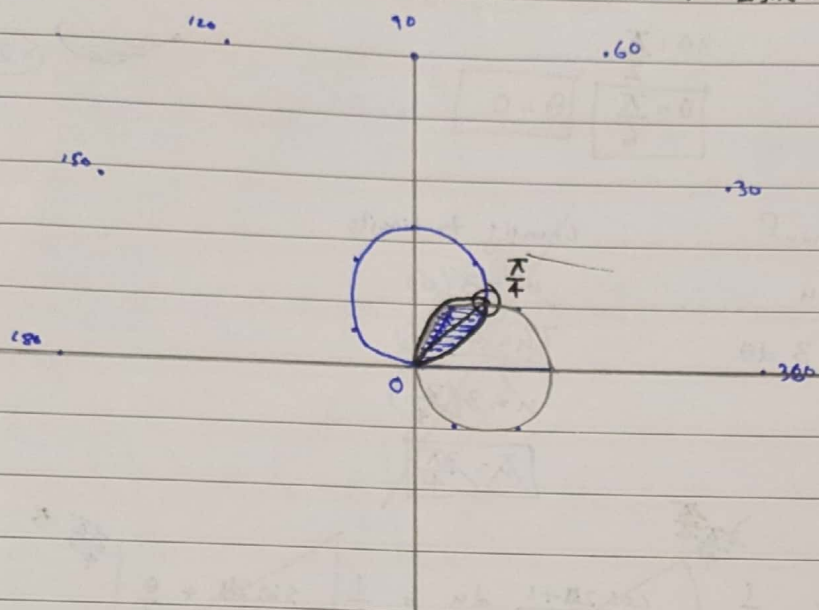
$$A = 2 \int_0^{\frac{\pi}{6}} \frac{1}{2} \cos^2 3\theta \, d\theta = \int_0^{\frac{\pi}{6}} \frac{\cos 6\theta + 1}{2} \, d\theta = \left[\frac{\sin 6\theta}{12} + \frac{\theta}{2} \right]_0^{\frac{\pi}{6}}$$

$$= \frac{\sin\left(6\left(\frac{\pi}{6}\right)\right)}{12} + \frac{\pi}{6} = \boxed{\frac{\pi}{6}} ???$$

Incorrect Answer

Day: _____

Date: _____

9. Shared by the circles $r = 2\cos\theta$ and $r = 2\sin\theta$ 

θ	$r = 2\cos\theta$	$r = 2\sin\theta$
0	2	0
30	$\frac{1.73}{\sqrt{3}}$	1
60	1	$\frac{1.73}{\sqrt{3}}$
90	0	2
120	-1	$\frac{1.73}{\sqrt{3}}$
150	$-\frac{1.73}{\sqrt{3}}$	1
180	-2	0
270	0	-2
360	2	0

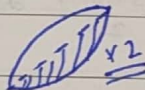
$$A = \int_0^{\frac{\pi}{4}} \frac{1}{2} (2\sin\theta)^2 d\theta + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{2} (2\cos\theta)^2 d\theta$$

$$2\cos\theta = 2\sin\theta$$

$$\tan\theta = 1$$

$$\theta = \tan^{-1}(1)$$

$$\theta = 45^\circ \text{ or } \frac{\pi}{4}$$

using symmetry 

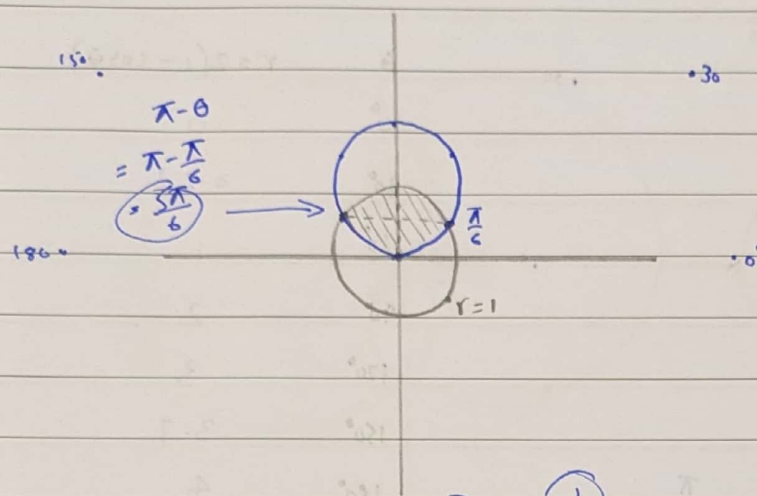
$$\therefore A = 2 \int_0^{\frac{\pi}{4}} \frac{1}{2} \cdot 4 \sin^2\theta d\theta = 2 \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= 2 \left| \theta - \frac{\sin 2\theta}{2} \right|_0^{\frac{\pi}{4}} = 2 \left| \frac{\pi}{4} - \frac{\sin(\frac{\pi}{2})}{2} \right| = 2 \left| \frac{\pi}{4} - \frac{1}{2} \right|$$

$$\boxed{2.57} \quad \boxed{= 0.57}$$

Day: _____

Date: _____

10. Shared by the circles $r=1$ and $r=2\sin\theta$ 

$$\theta \quad r=2\sin\theta$$

$$0^\circ \quad 0$$

$$30^\circ \quad 1$$

$$60^\circ \quad 1.73$$

$$90^\circ \quad 2$$

$$120^\circ \quad 1.73$$

$$150^\circ \quad 1$$

$$180^\circ \quad 0$$

$$A = \int_0^{\frac{\pi}{6}} \frac{1}{2} (2\sin\theta)^2 d\theta + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1}{2} (1)^2 d\theta \quad \theta = \sin^{-1} \frac{1}{2} \quad \boxed{\theta = \frac{\pi}{6}}$$

$$= \int_0^{\frac{\pi}{6}} 4\sin^2\theta d\theta = \int_0^{\frac{\pi}{6}} 2(1 - \cos 2\theta) d\theta = \int_0^{\frac{\pi}{6}} 2 - 2\cos 2\theta d\theta$$

$$= \left[2\theta - \frac{2\sin 2\theta}{2} \right]_0^{\frac{\pi}{6}} = \left[2\left(\frac{\pi}{6}\right) - \frac{2\sin 2\left(\frac{\pi}{6}\right)}{2} \right]$$

$$= \frac{0.36}{2} = 0.18 \quad \text{①} \Rightarrow \frac{1}{2} \left| \theta \right|_{\frac{\pi}{6}}^{\frac{\pi}{2}} =$$

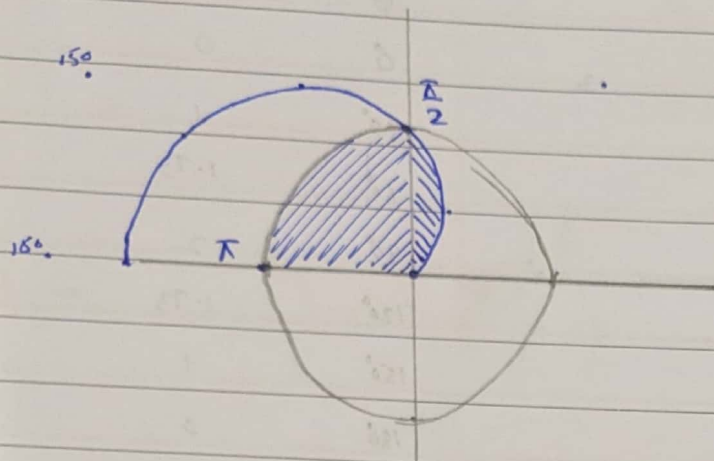
$$\left[\frac{2\left(\frac{\pi}{6}\right) - \frac{2\sin 2\left(\frac{\pi}{6}\right)}{2} \right] + \left[\frac{1}{2} \left| \frac{5\pi}{6} - \frac{\pi}{6} \right| \right]$$

$$\frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\pi}{3} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2} \approx \boxed{1.228} \checkmark$$

Day: _____

Date: _____

11. Shared by the circle $r = 2$ and the cardioid $r = 2(1 - \cos \theta)$.



$$\theta \quad r = 2(1 - \cos \theta)$$

0°	0
30°	0.27
60°	1
90°	2
120°	3
150°	3.7
180°	4

$$A = 2 \int_0^{\pi/2} \frac{1}{2} (2(1 - \cos \theta))^2 d\theta + 2 \int_{\pi/2}^{\pi} \frac{1}{2} (2)^2 d\theta$$

$$r = 2(1 - \cos \theta)$$

$$\cos \theta = 1$$

$$\theta = \cos^{-1}(1)$$

$$\theta = 0^\circ$$

$$= \int_0^{\pi/2} 4[1 - 2\cos \theta + \cos^2 \theta] d\theta + \int_{\pi/2}^{\pi} 4 d\theta$$

$$= \int_0^{\pi/2} 4 - 8\cos \theta + 4\left[\frac{\cos 2\theta + 1}{2}\right] d\theta + \left| 4\theta \right|_{\pi/2}^{\pi}$$

$$= \int_0^{\pi/2} 4 - 8\cos \theta + 2\cos 2\theta + 2 d\theta + \left| 4\theta \right|_{\pi/2}^{\pi}$$

$$= \left| 4\theta - 8\sin \theta + \frac{2\sin 2\theta}{2} + 2\theta \right|_0^{\pi/2} + \left| 4\theta \right|_{\pi/2}^{\pi}$$

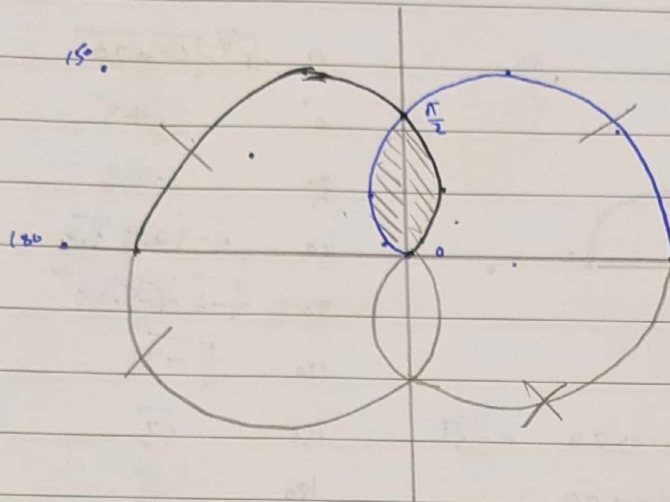
$$= \left[4\left(\frac{\pi}{2}\right) - 8\sin\left(\frac{\pi}{2}\right) + \sin\left(2\left(\frac{\pi}{2}\right)\right) + 2\left(\frac{\pi}{2}\right) \right] - \left[4(0) - 8\sin(0) + \sin(0) + 2(0) \right] + \left[4(\pi) - 4\left(\frac{\pi}{2}\right) \right]$$

$$= 3\pi - 8 + 0 + 2\pi$$

$$= 5\pi - 8 \approx 7.70 \quad \checkmark$$

Day: _____

Date: _____

12. Shared by the cardioids $r = 2(1 + \cos \theta)$ and $r = 2(1 - \cos \theta)$ 

θ	$r = 2(1 + \cos \theta)$	$r = 2(1 - \cos \theta)$
0°	4	0
30°	3.7	0.26
60°	3	1
90°	2	2
120°	1	3
150°	0.26	3.7
180°	0	4

$$A = \frac{1}{2} \int_0^{\pi/2} (2(1 - \cos \theta))^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [1 - 2\cos \theta + \cos^2 \theta] d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [1 - 2\cos \theta + \frac{\cos 2\theta + 1}{2}] d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [8 - 8\cos \theta + \frac{2\cos 2\theta + 2}{2}] d\theta$$

$$= \left[8\theta - 8\sin \theta + \frac{2\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$= \left[12\left(\frac{\pi}{2}\right) - 8\sin\left(\frac{\pi}{2}\right) + \frac{2\sin\left(2\left(\frac{\pi}{2}\right)\right)}{2} \right] - 0$$

$$= 6\pi - 8 + 0$$

$$= 6\pi - 8 = 6\pi - 16 \checkmark$$

$$r(1 + \cos \theta) = r(1 - \cos \theta)$$

$$1 + \cos \theta + \cos \theta = 1$$

$$2\cos \theta = 0$$

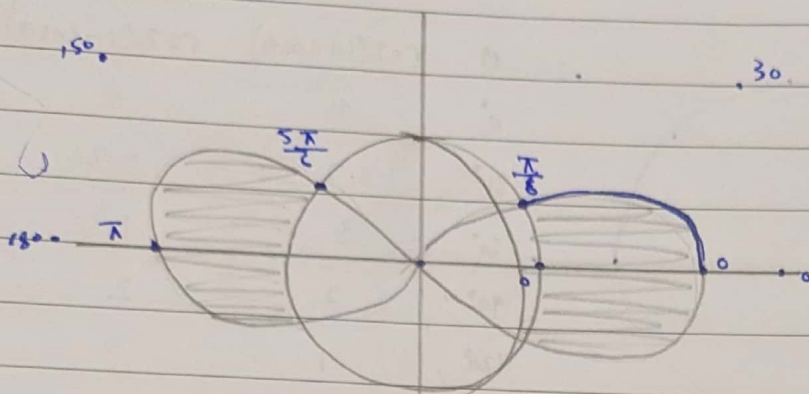
$$= \cos^{-1}(0)$$

$$\left(\frac{\pi}{2}\right)$$

Day: _____

Date: _____

13. Inside the lemniscate $r^2 = 6 \cos 2\theta$ and outside the circle $r = \sqrt{3}$



$$r^2 = 6 \cos 2\theta \quad r = \sqrt{3}$$

$$(\sqrt{3})^2 = 6 \cos 2\theta$$

$$2\theta = \cos^{-1} \frac{1}{2}$$

$$\theta = \frac{\pi}{3} \quad \approx \quad \boxed{\theta = \frac{\pi}{6}}$$

$$A = 4 \int_0^{\frac{\pi}{6}} \frac{1}{2} r^2 d\theta - 4 \int_0^{\frac{\pi}{6}} \frac{1}{2} r^2 d\theta$$

$$= 4 \int_0^{\frac{\pi}{6}} \left[\frac{1}{2} (\sqrt{6 \cos 2\theta})^2 - \frac{1}{2} (\sqrt{3})^2 \right] d\theta$$

$$= 4 \int_0^{\frac{\pi}{6}} 3 \cos 2\theta - \frac{3}{2} d\theta$$

$$= \int_0^{\frac{\pi}{6}} 12 \cos 2\theta - 6 d\theta$$

$$= \left| \frac{12 \sin 2\theta}{2} - 6\theta \right|_0^{\frac{\pi}{6}} = \left| 6 \sin \left[2 \left(\frac{\pi}{6} \right) \right] - 6 \left(\frac{\pi}{6} \right) \right| = \boxed{3\sqrt{3} - \pi}$$

$$\theta \quad \sqrt{r^2} = \sqrt{6 \cos 2\theta}$$

$$0 \quad 2.45$$

$$30 \quad 1.73$$

$$60 \quad \sqrt{-3} \rightarrow \sqrt{3}$$

$$90 \quad \sqrt{-6} \rightarrow \sqrt{-5.19}$$

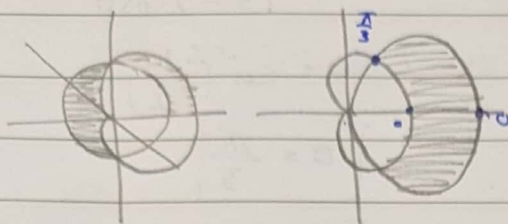
$$120 \quad \sqrt{-3}$$

$$150 \quad \sqrt{3}$$

$$180 \quad \sqrt{6}$$

Day: _____

Date: _____

14. Inside the circle $r = 3a \cos \theta$ and outside the cardioid $r = a(1 + \cos \theta)$, $a > 0$ 

$$3a \cos \theta = a(1 + \cos \theta)$$

$$2 \cos \theta = 1$$

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) = \boxed{\frac{\pi}{3}}$$

$$A = 2 \int_0^{\frac{\pi}{3}} \frac{1}{2} (3a \cos \theta)^2 d\theta - \frac{1}{2} [a(1 + \cos \theta)]^2 d\theta$$

$$= 2 \int_0^{\frac{\pi}{3}} \frac{1}{2} 9a^2 \cos^2 \theta - a^2 [1 + 2\cos \theta + \cos^2 \theta] d\theta$$

$$= a^2 \int_0^{\frac{\pi}{3}} \left[9 \left(\frac{\cos 2\theta + 1}{2} \right) - \left[1 + 2\cos \theta + \frac{\cos 2\theta + 1}{2} \right] \right] d\theta$$

$$= a^2 \left| \frac{9 \sin 2\theta}{4} + \frac{9\theta}{2} - \left(\theta + 2\sin \theta + \frac{\sin 2\theta}{4} + \frac{\theta}{2} \right) \right|_0^{\frac{\pi}{3}}$$

$$a^2 \left| \left[\frac{9\sqrt{3}}{8} + \frac{3\pi}{2} \right] - \left[\frac{\pi}{2} + \sqrt{3} + \frac{\sqrt{3}}{8} \right] \right|$$

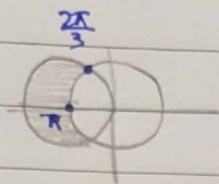
$$= \boxed{\pi a^2}$$

Incorrect
Answer

$$\boxed{a^2 (\pi + 1 - \sqrt{3})}$$

Day: _____

Date: _____

15. Inside the circle $r = -2\cos\theta$ and outside the circle $r = 1$ 

$$1 = -2\cos\theta$$

$$\theta = \cos^{-1}\left(-\frac{1}{2}\right)$$

$$\boxed{\theta = \frac{2\pi}{3}}$$

$$A = 2 \int_{\frac{2\pi}{3}}^{\pi} \frac{1}{2} (-2\cos\theta)^2 - \frac{1}{2} (1)^2 d\theta$$

$$= \int_{\frac{2\pi}{3}}^{\pi} 4\cos^2\theta - 1 d\theta = \int_{\frac{2\pi}{3}}^{\pi} 4\left(\frac{\cos 2\theta + 1}{2}\right) - 1 d\theta$$

$$= \left[4\left(\frac{\sin 2\theta}{4} + \frac{\theta}{2}\right) - \theta \right]_{\frac{2\pi}{3}}^{\pi}$$

$$= \left[4\left(\frac{\sin(2(\pi))}{4} + \frac{\pi}{2}\right) - \pi \right] - \left[4\left(\frac{\sin(2(\frac{2\pi}{3}))}{4} + \frac{(\frac{2\pi}{3})}{2}\right) - \frac{2\pi}{3} \right]$$

$$= \left[0 + \frac{2\pi}{2} - \pi \right] - \left[-\frac{\sqrt{3}}{2} + \frac{\pi}{3} - \frac{2\pi}{3} \right]$$

$$= 0 - \left(-\frac{\sqrt{3}}{2} - \frac{\pi}{3} \right)$$

$$= \pi + \frac{\pi}{3} + \frac{\sqrt{3}}{2}$$

$$\boxed{= \frac{4\pi}{3} + \frac{\sqrt{3}}{2}}$$

$$\therefore \boxed{\frac{\pi}{3} + \frac{\sqrt{3}}{2}} \checkmark$$

Day: _____

Date: _____

21. The spiral $r = \theta^2$, $0 \leq \theta \leq \sqrt{5}$

$$L = \int_0^{\sqrt{5}} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$\frac{dr}{d\theta} = 2\theta$$

$$\left(\frac{dr}{d\theta}\right)^2 = 4\theta^2$$

$$= \int_0^{\sqrt{5}} \sqrt{\theta^4 + 4\theta^2} d\theta$$

$$r = \theta^2$$

$$r^2 = \theta^4$$

$$= \int_0^{\sqrt{5}} \sqrt{\theta^4 + 4\theta^2} d\theta$$

$$= \int_0^{\sqrt{5}} \theta^2 + 2\theta d\theta$$

$$= \left| \frac{\theta^3}{3} + \frac{2\theta^2}{4} \right|_0^{\sqrt{5}} = \left| \frac{(\sqrt{5})^3}{3} + \frac{2(\sqrt{5})^2}{4} \right| = \boxed{6.22}$$

22. The spiral $r = \frac{e^\theta}{\sqrt{2}}$, $0 \leq \theta \leq \pi$

$$\frac{dr}{d\theta} = \frac{e^\theta}{\sqrt{2}}$$

$$= \int_0^\pi \sqrt{\left(\frac{e^\theta}{\sqrt{2}}\right)^2 + \left(\frac{e^\theta}{\sqrt{2}}\right)^2} d\theta$$

$$= \int_0^\pi \sqrt{2\left(\frac{e^\theta}{\sqrt{2}}\right)^2} d\theta$$

$$= \int_0^\pi \sqrt{2} \cdot \frac{e^\theta}{\sqrt{2}} d\theta$$

$$= \int_0^\pi \sqrt{2} \cdot \frac{1}{\sqrt{2}} \cdot e^\theta d\theta$$

$$= \sqrt{\frac{2}{2}} \int_0^\pi e^\theta d\theta = \sqrt{\frac{2}{2}} \left| e^\theta \right|_0^\pi = \boxed{24.28}$$

Day: _____

Date: _____

22. The spiral $r = e^\theta/\sqrt{2}$, $0 \leq \theta \leq \pi$

$$= \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$\left(\frac{dr}{d\theta}\right)^2 = \left(\frac{e^\theta}{\sqrt{2}}\right)^2$$

$$\boxed{\left(\frac{dr}{d\theta}\right)^2 = \frac{e^{2\theta}}{2}}$$

$$= \int_0^\pi \sqrt{\frac{e^{2\theta}}{2} + \frac{e^{2\theta}}{2}} d\theta$$

$$r = \frac{e^\theta}{\sqrt{2}}$$

$$\boxed{r^2 = \frac{e^{2\theta}}{2}}$$

$$= \int_0^\pi \sqrt{2\left(\frac{e^{2\theta}}{2}\right)} d\theta$$

$$= \int_0^\pi e^\theta d\theta$$

$$= \left| e^\theta \right|_0^\pi$$

$$= e^\pi - e^0$$

$$= \boxed{e^\pi - 1}$$

Day: _____

Date: _____

23. The cardioid $r = 1 + \cos \theta$

$$\frac{dr}{d\theta} = 1 + \cos \theta$$

$$= \int_0^{2\pi} \sqrt{\frac{3}{2} + 2\cos\theta + \frac{\cos 2\theta}{2}} d\theta \rightarrow -\frac{\sin\theta}{2} + \frac{\cos 2\theta}{2} d\theta$$

$$\frac{dr}{d\theta} = 0 \pm -\sin\theta$$

$$\left(\frac{dr}{d\theta}\right)^2 = -\sin^2\theta \rightarrow -\frac{1 + \cos 2\theta}{2}$$

$$= \int_0^{2\pi} \sqrt{1 + 2\cos\theta + \cos 2\theta} d\theta$$

$$r = 1 + \cos\theta$$

$$r^2 = 1 + 2\cos\theta + \cos^2\theta$$

$$= \int_0^{2\pi} 1 + \sqrt{2\cos\theta} + \sqrt{\cos 2\theta} d\theta$$

$$r^2 = 1 + 2\cos\theta + \left[\frac{\cos 2\theta}{2} + \frac{1}{2}\right]$$

$$\cos 2\theta = \cos^2\theta$$

$$\cos^2\theta = \frac{\cos 2\theta + 1}{2}$$

$$2\cos^2\theta \rightarrow \cos 2\theta + 1$$

Day: _____

Date: _____

24. The curve $r = a \sin^2\left(\frac{\theta}{2}\right)$, $0 \leq \theta \leq \pi$, $a > 0$

$$= \int_0^\pi \sqrt{\left(\frac{a - a \cos \theta}{2}\right)^2 + \frac{a^2 \sin^2 \theta}{4}} d\theta$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$\cos \theta = 1 - 2\sin^2\left(\frac{\theta}{2}\right)$$

$$= \int_0^\pi \sqrt{\frac{a^2}{4} - \frac{a^2 \cos^2 \theta}{4} + \frac{a^2 \sin^2 \theta}{4}} d\theta$$

$$\sin^2\left(\frac{\theta}{2}\right) = \frac{1 - \cos \theta}{2}$$

$$= \int_0^\pi \frac{a}{2} \sqrt{1 - \cos^2 \theta + \sin^2 \theta} d\theta$$

$$\frac{dr}{d\theta} = \frac{a - a \cos \theta}{2}$$

$$= 0 + \frac{a \sin \theta}{2}$$

$$= \frac{a}{2} \int_0^\pi \sqrt{1 - 1} d\theta$$

$$\left(\frac{dr}{d\theta}\right)^2 = \frac{a^2 \sin^2 \theta}{4}$$

$$\boxed{\frac{a}{2}}$$

Incorrect Answer

$$= \int_0^\pi \frac{a}{2} \left[\sqrt{1} - \sqrt{\cos^2 \theta} + \sqrt{\sin^2 \theta} \right] d\theta$$

$$= \frac{a}{2} \int_0^\pi 1 - \cos \theta + \sin \theta d\theta$$

$$= \frac{a}{2} \left[\theta - \sin \theta - \cos \theta \right]_0^\pi$$

$$= \frac{a}{2} \left[\pi - \sin \pi - \cos \pi \right] - \frac{a}{2} \left[0 - \sin(0) - \cos(0) \right]$$

$$= 2.07a + \frac{a}{2} \quad \frac{a}{2} \left[\pi + 1 \right] - \frac{a}{2} \left[-1 \right]$$

$$\frac{a}{2} \left[\pi + 1 + 1 \right]$$

$$= \frac{a}{2} [\pi + 2]$$

$$= \frac{a\pi}{2} + \frac{a2}{2}$$

Incorrect Answer

$$\boxed{\frac{\pi a}{2} + a}$$

Day: _____

Date: _____

25. The parabolic segment $r = 6/(1 + \cos \theta)$, $0 \leq \theta \leq \frac{\pi}{2}$

$$= \int_0^{\frac{\pi}{2}} \sqrt{\left[\frac{6}{(1+\cos \theta)}\right]^2 + \left[\frac{6 \sin \theta}{(1+\cos \theta)^2}\right]^2} d\theta$$

$$r = \frac{6}{(1 + \cos \theta)}$$

$$r^2 = \frac{36}{1 + 2\cos \theta + \cos^2 \theta}$$

~~$$= \int_1^2 \sqrt{\left(\frac{6}{u}\right)^2 + \left(\frac{6 \sin \theta}{u^2}\right)^2} \cdot \frac{1}{\sin \theta} du$$~~

$$\frac{dr}{d\theta} = 6 \cdot (1 + \cos \theta)^{-1}$$

$$= -6(1 + \cos \theta)^{-2} \cdot (-\sin \theta)$$

~~$$\frac{1}{\sin \theta}$$~~

$$\int_0^{\frac{\pi}{2}} \sqrt{\frac{36}{(1+\cos \theta)^2} + \frac{36 \sin^2 \theta}{(1+\cos \theta)^4}} d\theta$$

$$\frac{dr}{d\theta} = \Rightarrow \frac{6 \sin \theta}{(1 + \cos \theta)^2}$$

$$\int_0^{\frac{\pi}{2}} \sqrt{\frac{1}{(1+\cos \theta)^2} + \frac{36 + \frac{36 \sin^2 \theta}{(1+\cos \theta)^2}}{(1+\cos \theta)^2}} d\theta$$

$$\frac{1}{1+\cos \theta} \int_0^{\frac{\pi}{2}} 6 + \frac{6 \sin \theta}{1+\cos \theta} d\theta$$

~~$$1 + \cos \theta = u$$~~

~~$$\frac{du}{d\theta} = 0 - \sin \theta$$~~

~~$$\frac{1}{\sin \theta} du = d\theta$$~~

~~$$u = 1 + \cos(0) = 2$$~~

~~$$u = 1 + \cos\left(\frac{\pi}{2}\right) = 1$$~~



Day: _____

Date: _____

Exercise 11.6

Parabolas

Focus = ?

Sketch

Directrix = ?

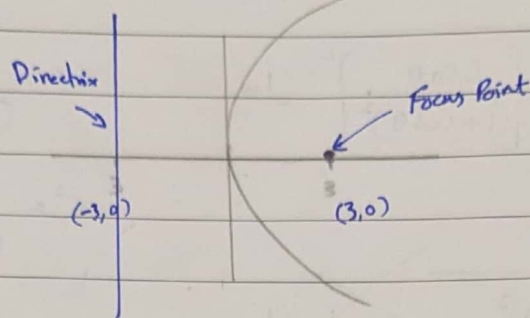
Equation = ?

9. $y^2 = 12x$

$y^2 = 4ax$

$y^2 = 4(3)x$

$a = 3$

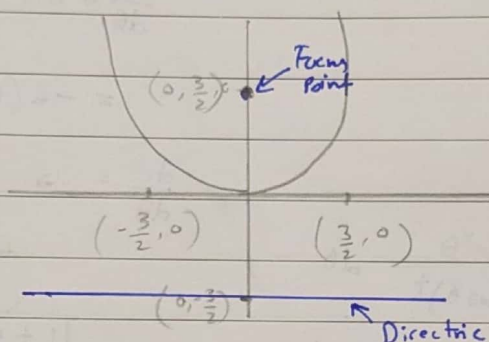


10. $x^2 = 6y$

$x^2 = 4ay$

$x^2 = 4(\frac{3}{2})y$

$a = \frac{3}{2}$

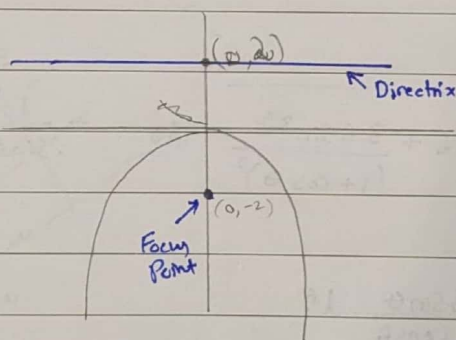


11. $x^2 = -8y$

$x^2 = -4ay$

$x^2 = -4(2)y$

$a = 2$

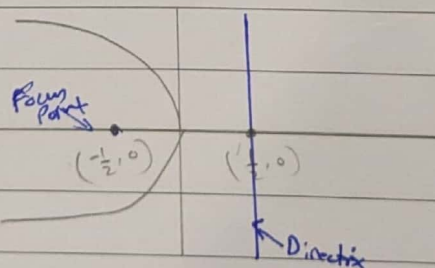


12. $y^2 = -2x$

$y^2 = -4ax$

$y^2 = -4(\frac{1}{2})x$

$a = \frac{1}{2}$



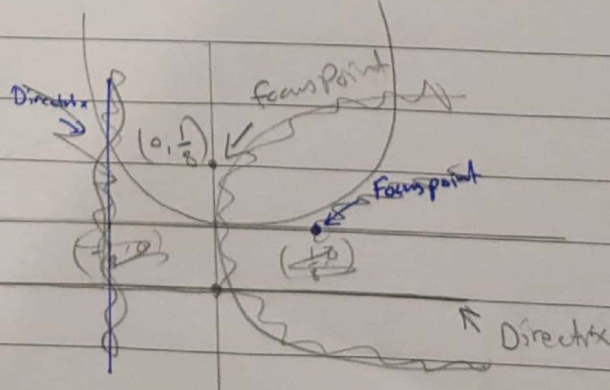
13. $y = 4x^2$

$x^2 = \frac{y}{4}$

$x^2 = 4ay$

$x^2 = 4(\frac{1}{8})y$

$a = \frac{1}{8}$



Day: _____

Date: _____

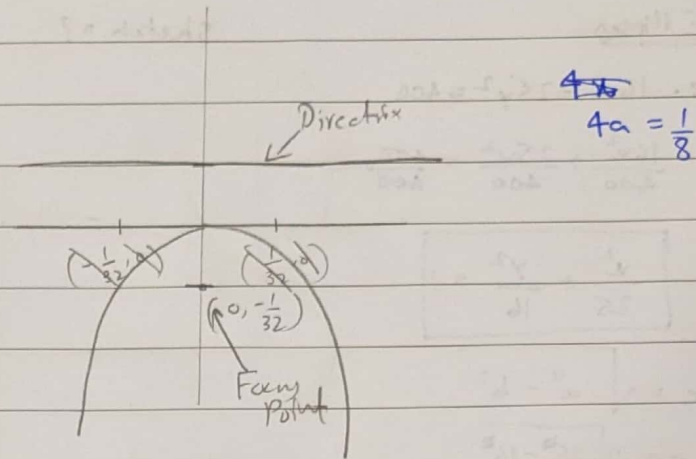
14. $y = -8x^2$

~~$-y \cdot \frac{1}{8} = x^2$~~

$x^2 = -\frac{1}{8}y$

$x^2 = -4ay$

$x^2 = -4\left(\frac{1}{32}\right)y$



15. $x = -3y^2$

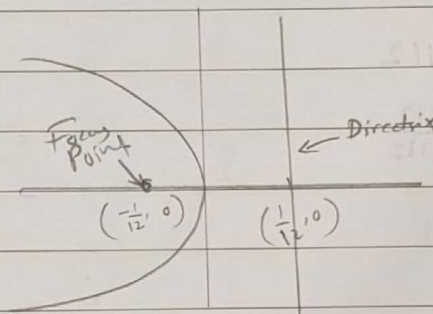
~~$-x = 3y^2$~~

$y^2 = -\frac{1}{3}x$

$y^2 = -4ax$

$x = y^2 = -4\left(\frac{1}{12}\right)x$

$a = \frac{1}{12}$



16. $x = 2y^2$

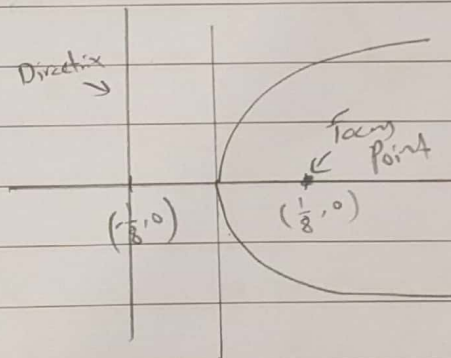
$\frac{1}{2}x = y^2$

$y^2 = \frac{1}{2}x$

$y^2 = 4ax$

$y^2 = 4\left(\frac{1}{8}\right)x$

$a = \frac{1}{8}$



Day: _____

Date: _____

Exercise 11.6Ellipses

Standard form = ? foci = ?

Sketch = ?

17. $16x^2 + 25y^2 = 400$

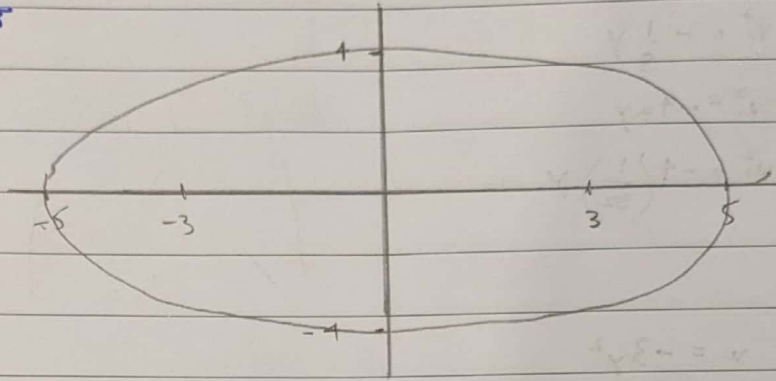
$$\frac{16x^2}{400} + \frac{25y^2}{400} = \frac{400}{400}$$

$$\boxed{\frac{x^2}{25} + \frac{y^2}{16} = 1}$$

$$c = \sqrt{a^2 - b^2}$$

$$c = \sqrt{25 - 16}$$

$$\boxed{c = 3}$$



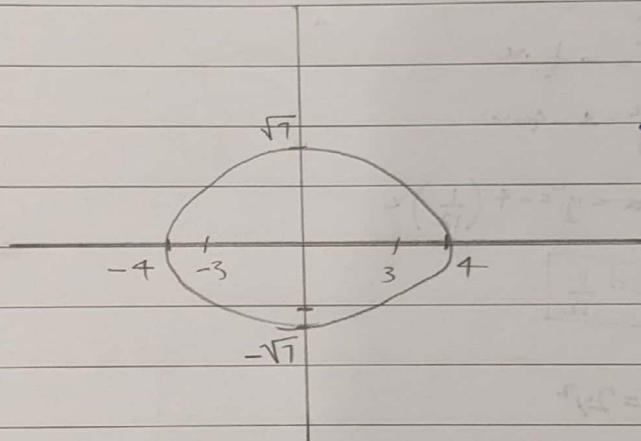
18. $7x^2 + 16y^2 = 112$

$$\frac{7x^2}{112} + \frac{16y^2}{112} = \frac{112}{112}$$

$$\frac{x^2}{16} + \frac{y^2}{7} = 1$$

$$c = \sqrt{16 - 7}$$

$$\boxed{c = 3}$$



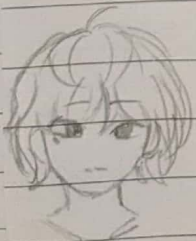
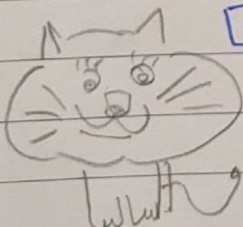
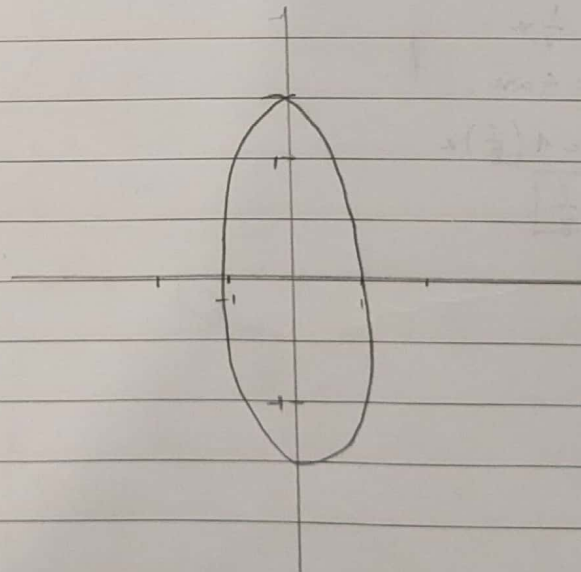
19. $2x^2 + y^2 = 2$

$$\frac{x^2}{1} + \frac{y^2}{2} = 1$$

$$c = \sqrt{1^2 - 2^2}$$

$$c = \sqrt{1 - 2} = \sqrt{-1}$$

$$\boxed{c = 1}$$



Day: _____

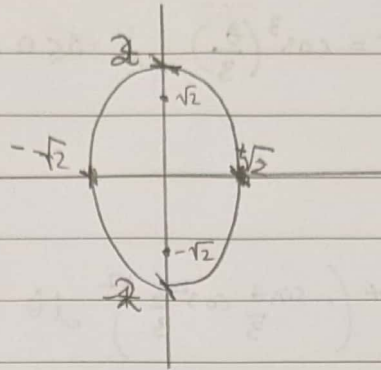
Date: _____

20. $2x^2 + y^2 = 4$

$$\frac{x^2}{2} + \frac{y^2}{4} = 1$$

$$c = \sqrt{4-2}$$

$$c = \sqrt{2}$$



21. $3x^2 + 2y^2 = 6$

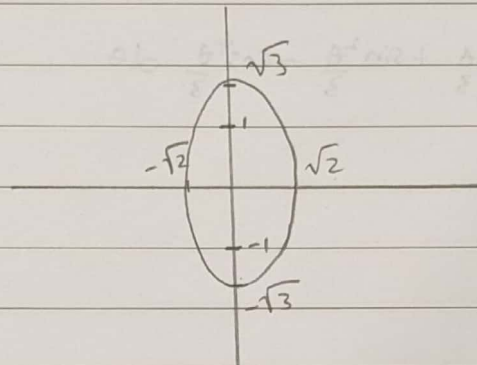
$$\frac{3x^2}{6} + \frac{2y^2}{6} = \frac{6}{6}$$

$$\frac{x^2}{2} + \frac{y^2}{3} = 1$$

$$c = \sqrt{3-2}$$

$$= \sqrt{1}$$

$$c = 1$$

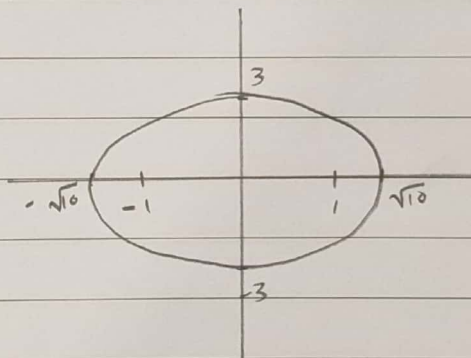


22. $x^2 + 10y^2 = 90$

$$\frac{x^2}{90} + \frac{y^2}{9} = 1$$

$$c = \sqrt{10-9}$$

$$c = 1$$

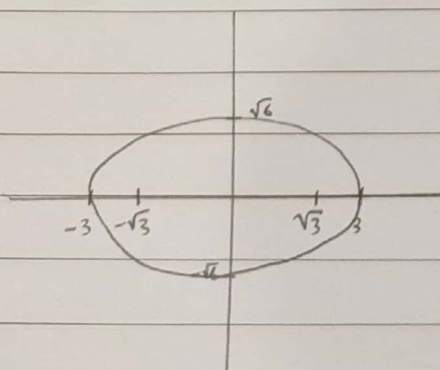


23. $6x^2 + 9y^2 = 54$

$$\frac{x^2}{9} + \frac{y^2}{6} = 1$$

$$c = \sqrt{9-6}$$

$$c = \sqrt{3}$$



24. $169x^2 + 25y^2 = 4225$

$$\frac{x^2}{25} + \frac{y^2}{169} = 1$$

$$c = \sqrt{169-25}$$

$$c = 12$$

